Estimating Phenylalanine Content of Foods from Ingredient Labels Using Machine Learning

Senior Capstone

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1 The Initial Problem

Phenylketonuria (PKU) is a rare genetic condition that impacts the body's ability to break down the amino acid phenylalanine when digesting food. Due to a deficiency of necessary liver enzymes, phenylalanine is not properly metabolized and instead accumulates in the bloodstream. Elevated levels of phenylalanine can lead to severe neurological and developmental issues, particularly if left unmanaged during early development.

Fortunately, the primary method for stopping the long-term risks associated with PKU is adherence to a carefully controlled low-protein diet. Individuals with PKU must closely track their daily phenylalanine intake using phenylalanine (Phe) estimates of foods that are calculated and maintained by trained dietitians within the flok organization and metabolic disorder community. Throughout the remainder of this report, phenylalanine will be referred to by its common abbreviation, Phe.

While the current system for estimating Phe content is generally sufficient for day-to-day management and allows individuals to maintain healthy lives, it still carries substantial margins of error that limit its precision. Additionally, the relationship between individual ingredients and total phenylalanine content is still a black box, largely due to a lack of regulation that could require this testing as a public health standard from food manufacturers. As a result, Phe estimation today relies heavily on expert rules and niche knowledge rather than reproducible, data-driven models.

The flok team is interested in exploring whether this manual estimation process can be augmented or replaced through machine learning and statistical modeling techniques. This approach would provide a parallel method for estimating Phe that does not rely on tribal knowledge passed down between dietitians, but instead on patterns learned from existing food data.

2 Proposed Solution

Our proposed solution is to identify mathematically discernible and meaningful relationships between the ingredient composition of a food item and its manually estimated Phe content. By using existing Phe estimates alongside ingredient lists and nutritional metadata, we aim to train machine learning models that can learn these relationships directly from data.

Ultimately, the goal is to develop a modeling system capable of producing Phe estimates comparable in accuracy to the existing manual process, while offering improved consistency and interpretability. Rather than relying on handcrafted rules and expert intuition alone, this approach seeks to ground Phe estimation in a reproducible statistical process.

3 Prototypes and Proof of Concept

3.1 Data Sources

We have been extremely fortunate to gain access to a rich and diverse dataset consisting of approximately 10,000 food items, each labeled with a total Phe estimation in mg/g. In addition to Phe values, the dataset includes detailed metadata such as ingredient lists and unrounded protein values. This data has been generated through decades of research and manual analysis and reflects current best practices in Phe estimation.

Before modeling, the ingredient lists were cleaned to reduce noise and improve consistency. Sub-ingredient groupings and non-ingredient phrases such as "contains less than 2% of" were removed. Foods with only one or two ingredients were excluded, leaving a final dataset of 6,855 foods. The dataset was initially split into training and testing sets using an 80/20 split across the full set of 6,855 food records. To reduce potential sampling bias and ensure more robust evaluation, we additionally validated model performance using cross-validation on the training set. The same data partitions were used across all experiments to ensure fair comparisons between models and feature sets. All models, including the least squares regression, regression forest, and baseline decision tree, were trained on the same training data with 3-fold cross validation, and evaluated on a consistent held-out test set.

4 Analyses

As an initial proof of concept, we started with an elementary baseline model built around a linear least-squares system. In this formulation, each food is represented as a weighted combination of its listed ingredients based on its rank in the ingredient list, and we attempt to solve for ingredient-level contributions that best explain the observed Phe values across the dataset. While intentionally simple, this baseline allows us to test whether there are meaningful, consistent relationships in the data before moving on to more complex or nonlinear modeling approaches.

4.1 Linear Baseline Model (Fall Semester)

The baseline model used in this project can be expressed as the following linear system:

$$Ax \approx b \tag{1}$$

where each component is defined as follows:

- $A \in R^{n\times m}$ is the ingredient composition matrix. Each row corresponds to a food item, and each column corresponds to a possible ingredient. The entry A_{ij} represents the relative contribution of ingredient j to food i, approximated using rank-based weights derived from the ingredient list and normalized so that each row sums to one.

- $x \in R^m$ is the vector of ingredient-level phenylalanine contributions. Each element x_j represents the estimated phenylalanine density (in mg of Phe per gram) associated with ingredient j.
- $b \in R^n$ is the target vector of observed phenylalanine values, where each entry b_i corresponds to the known phenylalanine content (in mg Phe per gram of food) for food i, as provided by the existing manual estimation process.

Solving this system via least squares yields an estimate of x that minimizes the total squared error between predicted and observed phenylalanine values across all foods. Although this formulation relies on simplifying assumptions about ingredient proportions and linearity, it provides a useful baseline for assessing whether ingredient-level structure in the data can explain observed Phe content.

We used Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Explained Variance (R^2) as our evaluation metrics for the baseline system, a standard that we maintain throughout the analysis of all subsequent models. When we first applied the model to the entire food dataset, the error was extremely high, which made it clear that a single global model was not a good fit for all foods (RMSE: 1.3071, MAE: 0.8376, R^2: 0.1806). This appeared to be driven by major structural differences between food groups, and pushed us to pivot toward testing the system on broader, ingredient-consistent subsets of the data.

To explore this, we applied five rough filters to the cleaned dataset and ran each resulting group through the linear system independently. These filters were designed to isolate foods with shared ingredient distributions and to test whether more homogeneous groups display more learnable relationships. For example, the common ingredient filter selects foods that share at least five frequently occurring ingredients, creating a large but more internally consistent subset. In contrast, the cheese filter isolates foods containing cheese or cheese-adjacent ingredients, producing a smaller but much more tightly defined group.

Food Group	# Foods	# Ingredients	RMSE	MAE	R^2
Common Ingredient Cluster	3999	282	1.2344	0.8150	0.1778
Cheeses / Cheese Alternatives	377	104	1.1671	0.6356	0.6365
Baked Goods	5235	287	1.2704	0.8381	0.1695
Protein Beverages	1090	188	1.1146	0.6075	0.4315
Savory Snacks	2843	254	1.1527	0.7596	0.2434

Table 1: Linear system performance across food groups (Fall Semester)

Cheese-based foods achieved the highest R^2, which implies a more linear and consistent relationship between ingredients and Phe content within this group. In contrast, baked goods and the common ingredient cluster performed the worst, likely due to their more heterogeneous and complex compositions of ingredients. Although the system was nearly full rank in all cases, indicating it was not underdetermined, the general poor fit suggests that model limitations are driven by nonlinear ingredient interactions rather than their positional ranking.

4.2 Random Forest (Spring Semester)

Observing the degree of nonlinearity between ingredients and Phe exposed by this initial study allowed us to pivot to a more nonlinear model. We chose to implement a random forest model, which performs well on tabular data and is much better at handling categorical structure and nonlinear interactions. With this approach, we hoped to capture relationships that were flattened or missed entirely by the linear least squares approach. This shift also allowed us to directly evaluate performance against our semester one baseline, where the linear model struggled to generalize across diverse food groups, and led to significant improvements in both predictive accuracy and overall model fit. Many of the issues observed in the fall, such as sensitivity to outliers and rigid assumptions about linearity, were mitigated by the increased capacity of the random forest. However, this added flexibility introduced new challenges, particularly in feature selection. Ingredient-based features, especially when represented through one-hot encoding, introduced significant noise that required careful handling, making feature engineering a central focus of this stage of the project.

4.3 Experiment

The experimental setup focused on evaluating how different feature engineering strategies impacted the performance of the regression forest model. A key component of this process was the construction of a new feature matrix that combined ingredient information and nutritional metadata, as well as the addition of engineered proxy variables derived from a supplementary dataset. This new dataset consisted of 164 common ingredients, along with their unrounded protein values per 100g, Phe per 100g, and corresponding Phe-to-protein ratios (multiplying factor). This allowed us to engineer new features that flag foods containing ingredients with MF >6 g/mg and unrounded protein >0.5 g. These thresholds were chosen to isolate ingredients that are most likely to contribute significantly to overall Phe content, based on domain knowledge from the supplementary dataset. Using these thresholds, we constructed features such as counts of high-MF ingredients, counts of high-protein ingredients, the ratio of high-protein ingredients to total ingredients, and average MF and protein values across matched ingredients. Our intention with this was to reintroduce ingredient information back into the model as meaningful signal rather than noise.

The tokenized and cleaned ingredient data from the main food dataset was additionally transformed into several representations, including binary indicators and lower-dimensional embeddings, to better capture structure while managing sparsity. In addition to basic binary ingredient indicators, we incorporated the engineered features described above to reflect both composition and rank-based importance within ingredient lists. To further address the high dimensionality of the ingredient space, truncated singular value decomposition (SVD) was applied to the ingredient matrix from the least squares approach across several component sizes. We ultimately selected `num_components = 60`, which produced a lower-dimensional representation that retained 56.1% of the variance while significantly improving model tractability. This value was selected as a balance between retaining sufficient variance from the original ingredient space and reducing dimensionality to improve model stability and generalization. Additionally, we introduced numeric features such as estimated protein density, calculated from protein content and serving weight, which proved to be a strong predictor of Phe content. Our final feature set consisted of 372 total features, including nutritional data, engineered features, and 60 low dimensional ingredient representations.

To provide a baseline for comparison, we implemented a `depth = 2` random forest as a heuristic Phe estimation function meant to mirror traditional rule-based methods.This heuristic approximates Phe content as a scaled function of protein, reflecting the historical approach used by those who track their Phe manually when new foods don't have lab estimations. This allowed us to directly compare learned models against established estimation practices.

To fully explore the capabilities of the regression forest we trained and evaluated it across multiple feature sets, including ingredient-only representations, protein and numeric features, and combined feature sets. This systematic comparison allowed us to isolate the contribution of each feature group and understand the trade-offs between model complexity and performance. The final model was selected based on its balance between predictive accuracy and generalization. It was trained using a combination of ingredient indicator features (where each ingredient is represented as a binary present/absent variable, later reduced via SVD), engineered ingredient statistics, supplementary dataset features, and numeric nutritional variables. Importantly, unrounded protein values were removed entirely from the feature set to better reflect real-world conditions, where such precise measurements are not available. This ensured that model performance more accurately represents its expected behavior in practical deployment.

5 Experimental Results

The final model was trained using a combination of numeric features, categorical metadata, SVD-reduced ingredient representations, and engineered MF-based features. The dataset was split using an 80/20 train-test split as stated before, resulting in 5,464 training samples and 1,367 test samples, with 3- fold cross validation ensuring consistency across all evaluated models.

To understand the contribution of different feature groups, we evaluated model performance across three configurations: ingredient-only features, protein and numeric features only, and the combined feature set (ingredients features, protein/numeric features, AND engineered features).

Model Variant	MAE Train	MAE Test	R^2 Train	R^2 Test
Combined Features	0.4069	0.5262	0.6414	0.4518
Ingredients Only	0.7996	0.8729	0.1125	0.0531
Only Protein Features	0.1279	0.1889	0.9107	0.7232

Table 2: Model performance across feature sets

These results highlight that ingredient-only features perform poorly in isolation, with near-zero ability to explain the variance in Phe values. In contrast, protein and numeric features alone produce the strongest performance, indicating that protein content remains the dominant predictor of Phe. The combined model improves significantly over ingredient-only features, but does not surpass the protein-only baseline, due to the removal of unrounded protein values as a feature to emulate real life application. This result highlights a central limitation of the approach, as it suggests that the model is not truly learning meaningful relationships from ingredient composition, but is instead relying primarily on protein as a proxy for Phe. To further analyze model behavior, we examined feature importance across the final random forest model.

While SVD-reduced ingredient features collectively contribute a substantial portion of total importance, their influence is distributed across many components rather than concentrated in interpretable signals. Numeric features, particularly protein-derived variables, remain the most impactful individual predictors. MF-based engineered features contribute minimally, suggesting limited coverage or weak alignment with the dataset. At the individual feature level, protein density emerges as the most important predictor (an engineered feature, followed by caloric and serving-based variables.

This reinforces the conclusion that the model primarily relies on protein-related and scale-related variables rather than extracting strong direct signals from ingredient composition.

Feature Type	Total Importance	Num Features	Top Feature
Numeric	0.2925	5	protein_density
Categorical	0.2779	302	measure_
Ingredients (SVD)	0.4214	60	svd_12
MF Features	0.0083	5	avg_mf

Table 3: Feature group importance, feature counts, and most important feature per group

6 Experiment Analysis

First looking at the performance of the final model, an R^2 of 0.45 shows a moderate fit to the data. The model seems to fail to capture the full relationships between ingredients, protein, and other numeric features, but still learns some degree of underlying structure. While nowhere near state-of-the-art, this amount of explained variance from a food label predictor aligns with existing papers attempting similar processes, highlighting the difficulty of learning from ingredient and nutritional features.

It beats out our baseline model by a significant margin, with the random forest reducing overall error and better capturing nonlinear structure in the data. It is important to note that the baseline model itself operates surprisingly close to the error margins typically seen in hand-calculated, "back-of-the-napkin" Phe estimations used in practice. As shown in Figure 1, both models exhibit similar central tendencies in their error distributions, with the random forest primarily improving on variance and reducing extreme outliers rather than completely shifting the baseline level of accuracy.

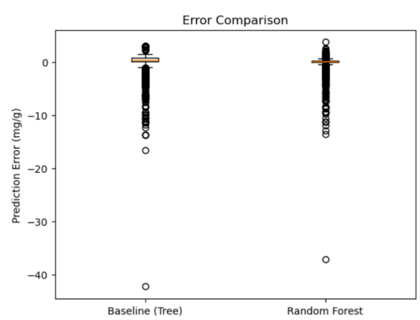

Figure 1: Prediction error distribution comparison

To get a full grasp of how well this model performs, we analyzed it within the context of standard consumption in the PKU community, primarily seeing how the model behaves as we scale food portions by size and Phe density.

We first examine model performance across Phe density buckets. As shown below, absolute error increases significantly as Phe density increases. Foods in the lowest range (0–50 mg/100g) exhibit relatively small absolute errors, while higher density foods show much larger deviations. The boxplot in Figure 2 highlights this clearly, with both the spread and upper tail of the error distribution growing as Phe density increases. This indicates that the model becomes less reliable as the underlying protein and Phe content increase, and struggles to capture the more complex nonlinear relationships present in high-Phe foods. However, this overall trend masks an important nuance: while the model overpredicts in the majority of cases, it systematically underpredicts in high-Phe foods, which are the most critical cases from a safety perspective.

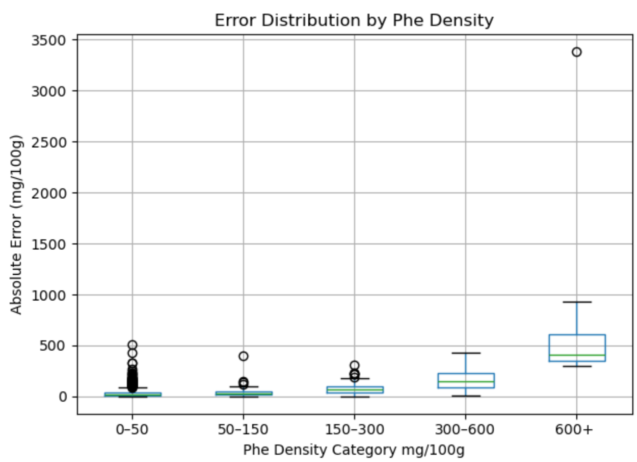

Figure 2: Error distribution across Phe density buckets

Looking more closely at the distribution, the variance in prediction error is not constant across the target range. Lower Phe foods are tightly clustered around low error values, while higher Phe foods exhibit large variance and extreme outliers. This suggests that the model is effectively smoothing predictions, capturing central tendencies while failing to accurately represent edge cases. In particular, very high Phe foods are consistently pulled downward toward the mean, resulting in large underestimation errors.

To better understand real-world impact, we scale these errors by portion size. Since Phe intake is measured in total milligrams rather than density, this provides a more practical evaluation. As expected, absolute error scales approximately linearly with portion size, meaning that even small per-gram errors can accumulate into large deviations for larger servings. While this behavior is mathematically straightforward, it has important implications in practice. Individuals tracking Phe intake make decisions based on total consumption, so larger portion sizes increase the consequences of prediction error. For example, a modest per-gram error may result in only minor deviations for a 50g portion, but can lead to substantial misestimation for a 250g portion. This is particularly concerning in high-Phe foods, where underestimation at larger serving sizes could lead to unsafe intake levels. As a result, model reliability becomes increasingly important as portion size increases, highlighting a significant limitation in real-world usage.

Metric	Value
Mean Average Error (mg/100g)	58.72
Median Absolute Error (mg/100g)	24.66
Mean Percent Error (%)	117.79
Avg Error @ 50g Portion (mg)	±29.36
Avg Error @ 100g Portion (mg)	±58.72
Avg Error @ 250g Portion (mg)	±146.80

Table 4: Error metrics scaled by food weight

Focusing specifically on a realistic consumption scenario, we isolate commonly logged snack-sized foods in the 20–40g range. Within this subset ($n = 393$), the model performs relatively well. The median absolute error is 33.69 mg, with a mean of 59.59 mg, and the majority of predictions fall within a manageable range. As shown in Figure 3, most predictions cluster near zero error, with only a small number of extreme outliers. This suggests that for smaller portion foods, which are common in daily consumption, the model behaves in a much more stable and predictable way.

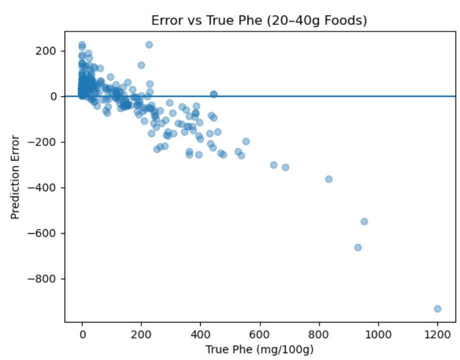

Figure 3: Prediction error versus true Phe content for snack-sized foods (20–40g)

We next examine prediction bias, specifically the tendency to overestimate versus underestimate Phe content. Across the full test set, the model overpredicts 78.79% of the time and underpredicts 21.21% of the time. This indicates a strong bias toward overestimation overall, which is ideal for model safety given this problem setting. When broken down by weight buckets, this trend remains consistent, with overprediction dominating across nearly all portion sizes.

Weight Range (g)	Overestimate (%)	Underestimate (%)
0–31	68.6	31.4
31–62	73.3	26.7
62–93	83.1	16.9
93–124	78.5	21.5
124–155	80.3	19.7
155–186	76.7	23.3
186–217	100.0	0.0
217–248	89.5	10.5

Table 5: Over vs. under prediction rates by food weight bucket

Within the snack-sized subset, this bias remains but is slightly reduced, with approximately 72.01% overprediction and 27.99% underprediction. This tendency toward overestimation can be considered relatively conservative in practice, as it errs on the side of caution. However, the underprediction that occurs in high-Phe foods remains more concerning, as it could lead to underestimating total intake.

Finally, we examine extreme errors. Without the per 100g scaling, only 6 out of 1367 test foods produced errors greater than 10 mg/g of Phe, representing a very small fraction of the dataset (0.0043%). This suggests that catastrophic failures are rare for pure Phe density predictions (mg/ gram of food). However, these failures are concentrated in high-Phe foods, reinforcing the earlier observation that the model struggles most in extreme regions of the target space. The worst prediction, for example, was a medical formula with extremely high Phe content, which prompted further inspection and attempts to filter out these types of outliers in pre-training. This unfortunately led to worse overall model performance, which is discussed in the following section.

Overall, several key takeaways emerge from this analysis. Primarily, the model behaves like a smoothed estimator, capturing general trends in the data while failing to represent extreme values. Despite this, it has shown it can perform reliably for low and moderate Phe foods, particularly in realistic portion sizes such as snack foods. However, it struggles with high-Phe foods, where nonlinear ingredient interactions become more important. This results in systematic bias, particularly under-prediction at high values, and limits the model's reliability in application.

7 Dead Ends

During the feature engineering phase of this project, the idea of using food brands as a feature was fully explored. Initially, the brands had an excessively high cardinality, blowing up our feature space with 3,000 noisy features. Instead of signaling brand-to-Phe trends and relationships to the model, the model failed to learn meaningful signal due to the large number of sparse, high-cardinality features introducing far more degrees of freedom than could be supported by the data. For the same reason of high cardinality, executing the random forest on only ingredients additionally led to complete model failure. In light of both of these errors, we decided to reduce the sparsity of both feature sets to allow the model learn lower dimensional representations. Using a character n-gram approach was the first operation tried on brands and ingredients. While this reduced the dimensionality of both feature sets to a maximum of 26x26 features, the model still failed to capture any signal over the noise from both feature groups. We dropped brands entirely as a feature, as it provided noise even in the best case, and pivoted to using truncated SVD for the ingredients.

One additional major dead end was using the serving size as a useful tool to train and evaluate the model. The serving size column was often inconsistent across products, with some units defined as "1 scoop" or "2 bars". This made the feature a strong source of noise and it did not reliably improve predictive performance on its own, or allow us to inspect per serving error metrics.

8 Lessons Learned

One of the main observations from this project is that model accuracy degrades as food size increases. While per-gram error may appear reasonable, this error scales linearly with portion size, leading to much larger absolute deviations in real-world settings. This is particularly important in the PKU context, where total Phe intake is measured in milligrams rather than concentration.

Another major observation was that feature engineering emerged as the primary bottleneck in this problem. While care was taken to ensure that the system was not underdetermined, this alone was not sufficient to achieve strong performance, as the main limitation lies in how ingredient information is represented rather than the capacity of the model. To address this, we attempted to engineer stronger signals by connecting ingredient data to known protein and Phe relationships through the supplementary MF dataset. Specifically, by mapping ingredients to their protein content and Phe-to-protein ratios in order to approximate underlying biochemical behavior, we allowed ourselves to reintroduce useful signal to the ingredients. This enabled the construction of features such as counts of high-protein and high-MF ingredients, normalized ratios, and average MF and protein values across matched ingredients. However, the impact of these features was limited in practice due to incomplete coverage and reliance on exact string matching, which resulted in many ingredients contributing no additional signal, ultimately reinforcing that the effectiveness of feature engineering in this setting is constrained by data alignment and representation quality.

More broadly, this project highlights the difficulty of working with ingredient data. Raw ingredient lists do not translate cleanly into structured features, and much of the meaningful signal is either implicit or lost during preprocessing. Capturing these relationships in a way that a model can learn from remains a core challenge.

Finally, model performance must be evaluated in the context of risk within the PKU community. Small errors in low-Phe foods are generally acceptable, but larger errors in high-Phe foods carry significantly greater risk. In this setting, absolute error is more important than relative error, especially as portion sizes increase. As a result, the model is best positioned as a supporting tool to assist dietitians, rather than a standalone decision-making system.

9 Future Work

One natural extension of this work would be to reframe the problem as a classification task, predicting whether a food falls "higher" or "lower" than the 50mg of Phe $\times$ rounded protein baseline. This would better align with how dietitians make quick decisions in practice, where exact values are less critical than whether a food crosses a meaningful threshold. By reducing the sensitivity to exact numerical error, this approach could potentially produce more stable and usable outputs, especially in cases where regression models struggle with extreme values.

Another direction for future work is shifting the focus of the model from point prediction to prediction confidence. Rather than forcing the model to output a value for every food, a confidence-aware system would allow it to identify when it is operating within a well-understood region of the data versus when it is extrapolating. This builds directly on earlier observations that certain foods and ingredient combinations produce consistently stable predictions, while others introduce high variance. In practice, a system that can abstain or flag low-confidence predictions would be significantly more useful, as it aligns with the real-world requirement of knowing when a model should not be trusted.

10 Project Context

This model is intended to be used internally by the in-organization dietitian, as an additional point of comparison rather than a source of truth. The goal is not to replace their estimation process, but to support it. For example, once the dietitian completes a Phe estimation for a food, the model can be used to sanity-check the result. If the two values are close, it provides confidence; if they differ significantly, it may surface a mistake or prompt a second look at the assumptions used.

Small errors in Phe estimation are already part of managing PKU. Blood Phe levels are monitored on a monthly basis, and regulation happens over a long time window. Because of this, a single food with a slightly misreported Phe value is not immediately harmful for those with PKU, which makes this model a safe and useful tool for reducing long-term estimation error rather than needing perfect precision.

11 Ethical and Privacy Concerns

All data used in this project is food-level nutritional data and contains no personal health information, minimizing direct privacy risks for those that use and rely on flok data. The dataset reflects expert dietitian estimates rather, not individual patient records, ensuring no identifiable user data is exposed or modeled. In addition, access to the data and resulting models is restricted to internal flok team members, preventing misuse or misinterpretation by non-experts.

12 Resources

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